Recitation Week 10

Hi, my name is Greed and I am one of the emotions that reside in your brain. I love money. Who says money can't buy happiness? In fact, I have infinite amount of money. Unfortunately, I only have coins in different denominations. I often wonder... what is the best way to pay people in coins? Can you help me to spend some of my money?

Consider the problem of making changes of n cents using the fewest number of coins. The coins come from $m+1$ different denominations $d_0, \ldots, d_m$.

1. Consider an optimal solution to the problem of making change for n cents. Let $C(n)$ be the optimal number of coins to make change for n cents. Select an arbitrary coin from the optimal solution. Assume that the denomination of the coin is d_k. Argue that $C(n) = 1 + C(n - d_k)$.

   Note that this implies that if we are able to select and remove a coin that belongs to the optimal solution, we are left with a subproblem of optimally making change for remaining amount (optimal substructure).

2. Suppose that the available coins are in denominations that are powers of c, i.e. the denominations are $c^0, c^1, c^2, \cdots, c^m$ for $c \geq 2$. We first prove a fact that is useful for later showing the optimality of our algorithm: In the optimal solution, the number of coins in each denomination c^k for $k = 0, 1, \ldots, m-1$ (all except the largest denomination) is at most $c - 1$.

3. Now show the following: If $c^k \leq n \leq c^{k+1}$ for $k < m$, the optimal solution must contain a coin from denomination c^k. Also, show that if $n \geq c^m$, the optimal solution must contain a coin from denomination c^m.

   (Hint: Using the result in the previous part, what is the largest number that can be made up using coins from the optimal solution that comes from denominations $c^0, \ldots, c^k$.)

4. We have shown a greedy choice property that corresponds to the optimal substructure property for this problem, i.e. the optimal solution always contains a coin of the largest denomination that is less than or equal to n.

   Using the result, describe a greedy algorithm to optimally making change for n cents from denomination $c^0, \ldots, c^m$. Give pseudo-code.